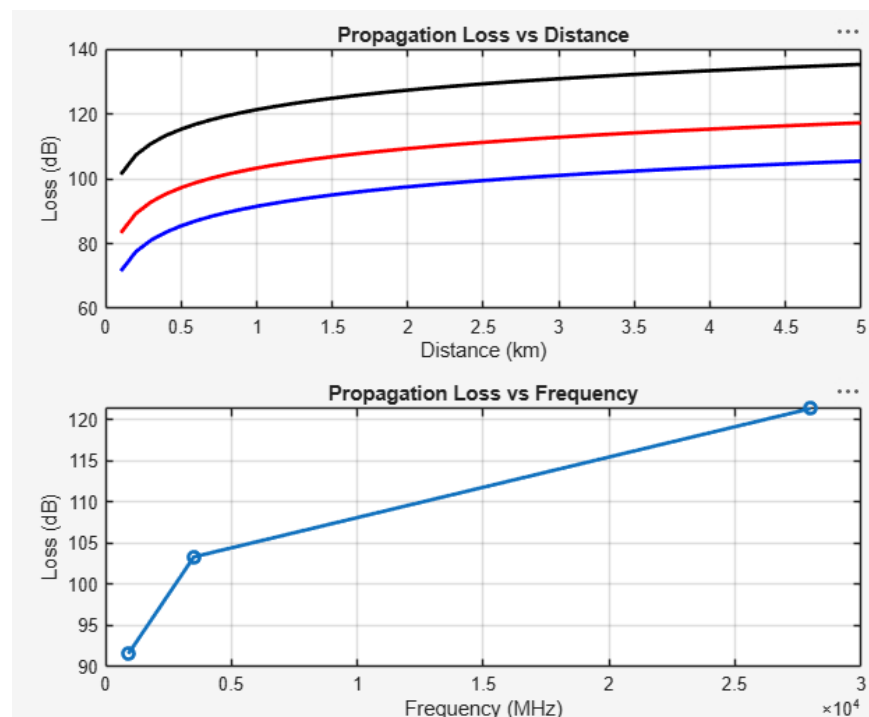


Ex.N.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation

Objective 1: Matlab Code and Output

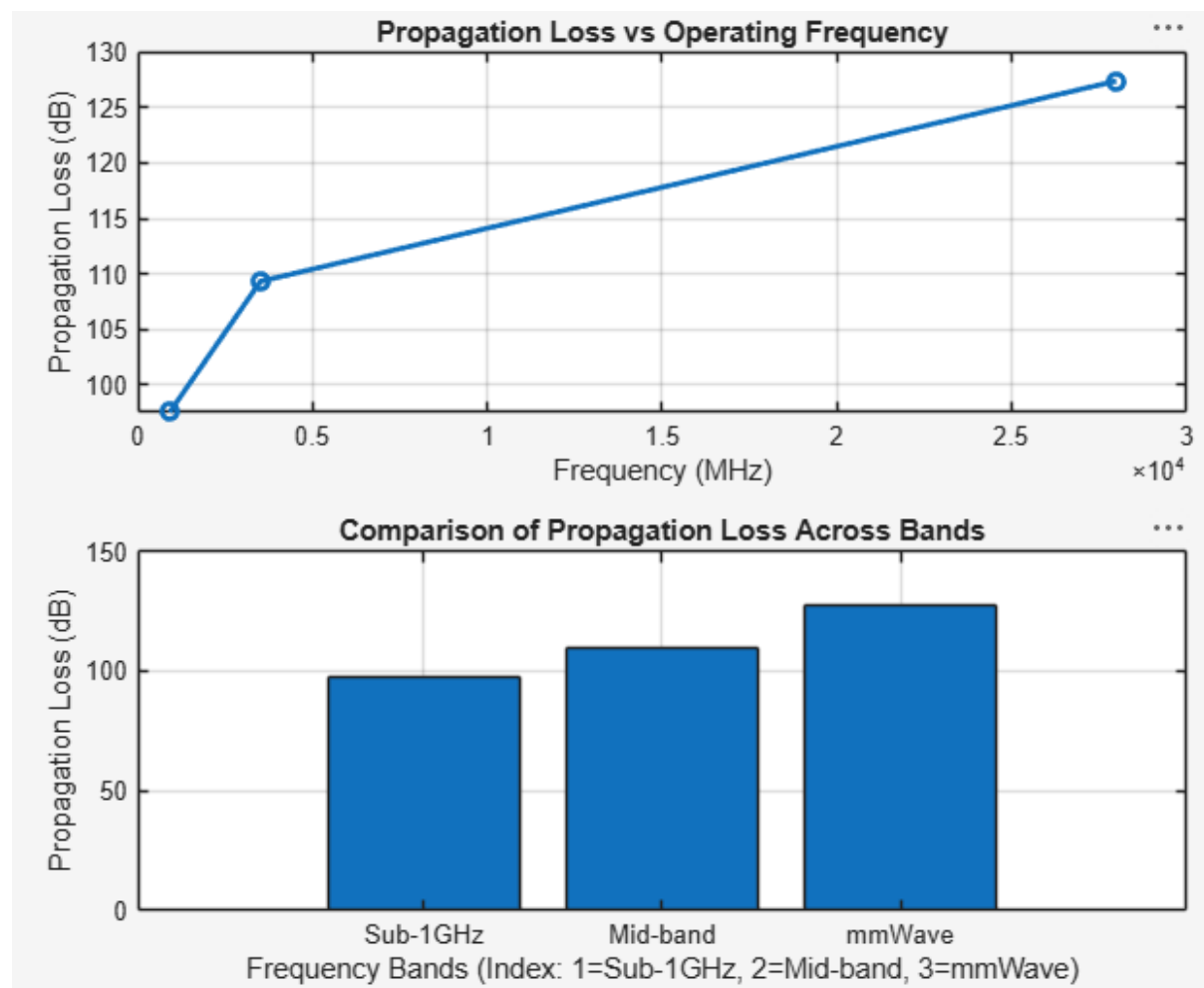
/MATLAB Drive/slotb3.m

```
1  clc; clear; close all;
2  d = 0.1:0.1:5; % Distance (km)
3  f = [900 3500 28000]; % Frequency (MHz)
4  PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
5  PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
6  PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
7  figure
8  % ----- Graph 1 -----
9  subplot(2,1,1)
10
11  plot(d,PL1,'b','LineWidth',2); hold on
12  plot(d,PL2,'r','LineWidth',2)
13  plot(d,PL3,'k','LineWidth',2)
14  title('Propagation Loss vs Distance')
15  xlabel('Distance (km)')
16  ylabel('Loss (dB)')
17  grid on
18  % ----- Graph 2 (NO legend used to avoid error)
19  subplot(2,1,2)
20  Freq = [900 3500 28000];
21  Loss = [PL1(10) PL2(10) PL3(10)];
22  plot(Freq,Loss,'-o','LineWidth',2)
23  title('Propagation Loss vs Frequency')
24  xlabel('Frequency (MHz)')
25  ylabel('Loss (dB)')
26  grid on
27
```



Objective 2: Matlab Code and Output

```
28     d = 2; % fixed distance (km)
29
30     f = [900 3500 28000]; % MHz
31     PL = 32.44 + 20*log10(f) + 20*log10(d);
32     figure
33     % ----- Graph 1: Frequency vs Loss -----
34     subplot(2,1,1)
35     plot(f,PL,'-o','LineWidth',2)
36     title('Propagation Loss vs Operating Frequency')
37     xlabel('Frequency (MHz)')
38     ylabel('Propagation Loss (dB)')
39     grid on
40     % ----- Graph 2: Band Comparison -----
41     subplot(2,1,2)
42     bar(PL)
43     title('Comparison of Propagation Loss Across Bands')
44     xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
45     ylabel('Propagation Loss (dB)')
46     xticklabels({'Sub-1GHz','Mid-band','mmWave'})
47     grid on
48
```



Objective 3: Matlab Code and Output

```
49 % Frequency bands (MHz)
50 f = [900 3500 28000];
51 % Distance (km)
52 d = 2;
53 % Propagation Loss (FSPL)
54 PL = 32.44 + 20*log10(f) + 20*log10(d);
55 figure
56 % ----- Graph 1: Simple Line Plot (SAFE) -----
57 subplot(2,1,1)
58 plot(f, PL, '-o', 'LineWidth', 2)
59 title('Propagation Loss vs Operating Frequency')
60 xlabel('Frequency (MHz)')
61 ylabel('Propagation Loss (dB)')
62 grid on
63 % ----- Graph 2: Safe Comparison Plot -----
64 subplot(2,1,2)
65 plot(1:3, PL, '-s', 'LineWidth', 2)
66 title('Band-wise Propagation Loss Comparison')
67 xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
68 ylabel('Propagation Loss (dB)')
69 grid on
```

